

Question Paper

Exam Date & Time: 22-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 2123/ECE_2123 SIGNALS & SYSTEMS

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

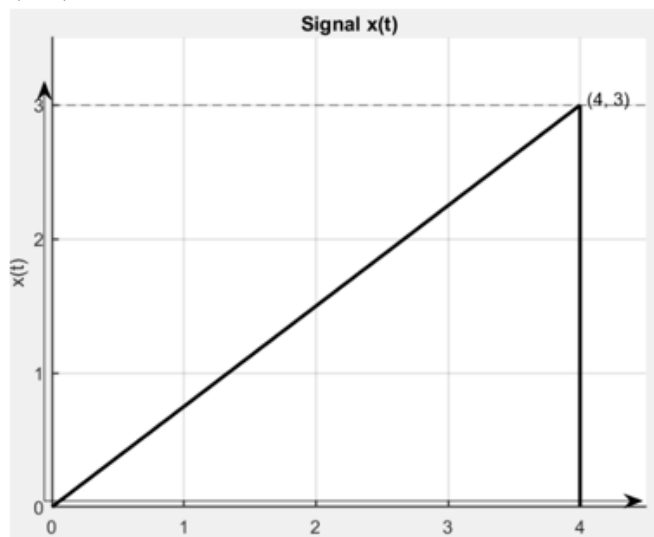
Answer all the questions.

- 1A) Determine the Fourier transform of the signal and plot magnitude and phase spectrum (5)

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Sketch the following signal for $x(t)$ shown in figure (3)
 $x(3t+2)$
 $x(-2t-1)$



Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Determine whether the following signal is periodic- if periodic, determine the fundamental period. (2)
 $x(t) = \cos(2t) + \sin(3t)$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) For each of the following continuous-time systems, analyze and state whether the system is **linear**, **has memory**, and is **causal**. (5)

a) $y(t) = \frac{1}{t}x(t)$

b) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

- 2B) Find Z-transform of the following signal and plot poles, zeros and RoC (3)

$$x[n] = \left(\frac{1}{5}\right)^n u[-n]$$

- 2C) Draw direct form I implementation of the system- (2)

$$y(t) + 3\frac{dy(t)}{dt} = 2x(t) + 4\frac{dx(t)}{dt}$$

- 3A) Determine the DTFT of the signal, (5)

$$x[n] = n \left(\frac{1}{2}\right)^n u[n - 3]$$

- 3B) Consider a Discrete Time signal (3)

$$x[n] = u[n + 3] - 2u[n] + u[n - 4]u[n+3].$$

Sketch

a) $x[n]$

b) $x[2n - 3]$

c) $-x[3 - n]$

- 3C) Illustrate the product and shifting property of discrete impulse function. (2)

- 4A) For each impulse response, determine whether the corresponding LTI system is (5)

- (i) Memory less
- (ii) Causal
- (iii) Stable

$$h_1[n] = (0.8)^{|n|} \text{ and } h_2[n] = (0.7)^n u[n] - (0.7)^{n-4} u[n - 4]$$

- 4B) Draw direct form I and direct form II implementations for the following difference equation (3)

$$y[n] + \frac{1}{2} y[n - 1] - y[n - 3] = 3 x[n - 1] + 2 x[n - 2]$$

- 4C) Evaluate the discrete-time convolution sum (2)

$$y[n] = u[n] * u[n - 3]$$

- 5A) For each of the following signals, sampled with sampling interval T_s , determine the bounds on T_s , which guarantee that there will be no aliasing: (5)

$$x(t) = \frac{1}{t} \sin(3\pi t) + \cos(3\pi t)$$

$$x(t) = e^{-6t}u(t) * \left(\frac{\sin Wt}{\pi t}\right)$$

Course Outcome: "CO5"

5B) For the following system, determine whether is memoryless, stable, causal, linear and time invariant: (3)

$$y[n] = x[n] \sum_{k=-\infty}^{\infty} \delta[n - 2k]$$

Course Outcome: "CO4"

5C) Consider a signal $x(t) = \left(\frac{\sin 2\pi t}{\pi t}\right)$. Sketch the frequency spectrum of the signal when (2)
sampled at two different intervals: $T_s = 0.125$ seconds and $T_s = 0.5$ seconds.

Course Outcome: "CO5"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2123:SIGNALS AND SYSTEMS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	8
5	5	CO 5	7
Total			45

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2123:SIGNALS AND SYSTEMS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	27(90%)
4	Analysis	1	3(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		9	30 (100%)